MATHEMATICS PART 1 (ALGEBRA) | CLASS 10 | MAHARASHTRA SSC BOARD

Chapter 6: Statistics

MODEL ANSWER SHEET / ANSWER KEY

38	4	11
QUESTIONS	IN TEXTBOOK	FIGURES

HOW TO USE
Cover the answer box ✓ and answer first. A **TEXTBOOK** *strip means the same question is in your Balbharati Math-1 textbook, at the set and page shown. Tick ☐ each section once revised.*

Q.1 (A)	Q.1 (B)	Q.2 (A)	Q.2 (B)	Q.3 (A)	Q.3 (B)	Q.4	Q.5
Multiple Choice	One-step Solve	Activities	Short Solve	Activities	Solve / Prove	Long Solve / Prove	Creative / HOTS
10 Q	6 Q	4 Q	3 Q	4 Q	3 Q	5 Q	3 Q

Q.1 (A) Multiple Choice — 1 Mark each

10 QUESTIONS ☐ Revised

Q1 Following table shows the percentage of donors as per the blood group, the central angle for the group 'O' is:

Blood group	O	A	B	AB
% Of persons	60	20	15	5

(a) 72° (b) 108° (c) 216° (d) 54°

ANSWER / SOLUTION

216°

Explanation:

The central angle for a sector $= \frac{\text{Percentage}}{100} \times 360^\circ$.

So for O: $\frac{60}{100} \times 360^\circ = 0.6 \times 360^\circ = 216^\circ$

Q2 If the following information that is budget of a family is shown as a pie graph, then central angle for clothing is:

Head	Rent	Clothing	Eduaction	Saving	Food	Miscellaneous
Expenditure	Rs. 2400	Rs. 1800	Rs. 1200	Rs. 1200	Rs. 4800	Rs. 600

(a) 72° (b) 54° (c) 36° (d) 18°

ANSWER / SOLUTION

54°

Explanation:

Total expenditure $= 2400 + 1800 + 1200 + 1200 + 4800 + 600 = \text{Rs. } 12,000$

Clothing $= \text{Rs. } 1800$

Fraction $= \frac{1800}{12000} = 0.15$

Central angle $= 0.15 \times 360^\circ = 54^\circ$

Q3 The class mark of the class $15 - 20$ is _____.

(a) 17.5 (b) 15.5 (c) 17 (d) 18.5

✓ **ANSWER / SOLUTION**

The class mark of the class $15 - 20$ is **17.5**.

Explanation:

The class mark (midpoint) $= \frac{\text{Lower limit}+\text{Upper limit}}{2} = \frac{15+20}{2} = 17.5$

Q4 If assumed mean $(A) = 57.5$, $\sum f_i d_i = -10$ and $\sum f_i = 80$ then Mean = _____.

(a) 57.625 (b) 57.125 **(c) 57.375** (d) 57.275

✓ **ANSWER / SOLUTION**

If assumed mean $(A) = 57.5$, $\sum f_i d_i = -10$ and $\sum f_i = 80$ then Mean = **57.375**.

Explanation:

Using the assumed-mean formula $\bar{X} = A + \frac{\sum f_i d_i}{\sum f_i}$.

Compute: $\frac{\sum f_i d_i}{\sum f_i} = \frac{-10}{80} = -0.125$

So, mean $= 57.5 + (-0.125) = 57.375$

Q5 Which of the following is not a measure of central tendency?

(a) Median **(b) Deviation** (c) Mode (d) Mean

✓ **ANSWER / SOLUTION**

Deviation

Explanation:

Mean, median and mode are measures of central tendency. Deviation measures dispersion/spread around that center, not the center itself.

Q6 If the mean of frequency distribution is 34.9, $\sum f_i x_i = 1000 + a$, $\sum f_i = 30$ then the value of a = _____.

(a) 47 (b) 48 (c) 42 (d) 45

✓ **ANSWER / SOLUTION**

If the mean of frequency distribution is 34.9, $\sum f_i x_i = 1000 + a$, $\sum f_i = 30$ then the value of a = **47**.

Explanation:

$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$

So, $\sum f_i x_i = 34.9 \times 30 = 1047$

Given $\sum f_i x_i = 1000 + a$,

We have $1000 + a = 1047 \implies a = 47$

Q7 A box contains nails of different lengths as shown below, the median class will be:

Length (in cm)	2.0 – 2.5	2.5 – 3.0	3.0 – 3.5	3.5 – 4.0	4.0 – 4.5
No. of nails	5	18	7	11	9

(a) 2.5 – 3.0 **(b) 3.0 – 3.5** (c) 3.5 – 4.0 (d) 4.0 – 4.5

✓ **ANSWER / SOLUTION**

3.0 – 3.5

Explanation:

To determine the median class, we first need to construct a cumulative frequency table:

Q7 *continued...*

ANSWER / SOLUTION (CONTD.)

Length (in cm)	No. of nails (f)	Cumulative Frequency (cf)
2.0 – 2.5	5	5
2.5 – 3.0	18	$5 + 18 = 23$
3.0 – 3.5	7	$23 + 7 = 30$
3.5 – 4.0	11	$30 + 11 = 41$
4.0 – 4.5	9	$41 + 9 = 50$

1. Find the total frequency (N)

Total number of nails (N) $= 5 + 18 + 7 + 11 + 9 = 50$

2. Calculate $\frac{N}{2} = \frac{50}{2} = 25$

3. Identify the median class

Look for the first cumulative frequency that is greater than or equal to 25.

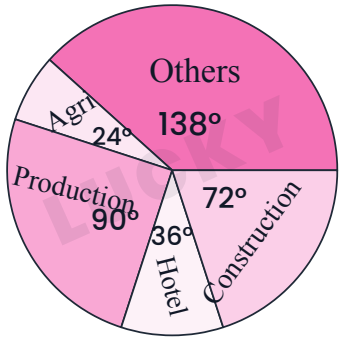
The cumulative frequency for the class interval 2.5 – 3.0 is 23 (which is less than 25).

The cumulative frequency for the next class interval 3.0 – 3.5 is 30 (which is greater than or equal to 25).

Thus, the median value lies within the interval 3.0 – 3.5.

Q8 **Look at the pie diagram showing number of skilled workers in different fields the number of workers in the production is ---------- if total number of workers is 10,000**

(a) 2000 (b) 1500 (c) **2500** (d) 3500



ANSWER / SOLUTION

2500

Explanation:

The production sector's central angle is $90°$.

So, its fraction of the circle is $\frac{90°}{360°} = \frac{1}{4}$.

Therefore, $\frac{1}{4}$ of $10,000 = 10,000 \times 0.25 = 2,500$.

Q9 **Class marks in a grouped frequency table are useful to find:**

(a) **Mean** (b) Median (c) Mode (d) All of these

ANSWER / SOLUTION

Mean

Explanation:

Class Mark (x_i): This is the midpoint or central value of a class interval, calculated as:

$\text{Class Mark} = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$

Finding the Mean: When calculating the mean ($\bar{x}$) for grouped data, we assume that the frequency of each class is centered around its midpoint. The direct formula explicitly uses the class marks (x_i):

$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

Median and Mode: Calculating the median and mode of grouped data relies on class boundaries, cumulative frequencies or individual class frequencies, rather than the class marks.

Q10 The above data is to be shown by a frequency polygon. The coordinates of the points to show number of students in the class $130 - 140$ are:

Heights (in cm)	100 - 110	110 - 120	120 - 130	130 - 140	140 - 150
No. of students	6	8	14	10	4

(a) $(130, 10)$
(b) $(135, 10)$
(c) $(10, 135)$
(d) $(10, 130)$

ANSWER / SOLUTION

(135, 10)

Explanation:

For a frequency polygon each point is (class midpoint, frequency).

Midpoint of $130 - 140$ is $\frac{130+140}{2} = 135$ and the frequency is 10.

So, the coordinate is $(135, 10)$.

Q.1 (B) One-step Solve 1 Mark each

6 QUESTIONS · 1 ☐ Revised

Q1 Find the class mark of the class $55 - 60$.

ANSWER / SOLUTION

Given: Class interval $55 - 60$

Step-wise calculation:

Lower limit $(L) = 55$

Upper limit $(U) = 60$

$\text{Class mark} = \frac{\text{Lower limit} + \text{Upper limit}}{2} = \frac{55+60}{2} = \frac{115}{2} = 57.5$

The class mark of $55 - 60$ is 57.5.

Q2 Write the Modal class

Class interval	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35
Frequency	30	35	40	75	15

ANSWER / SOLUTION

1. Analyze the given frequency table

To find the modal class, arrange the data into its corresponding pairs:

Class Interval	Frequency
10 – 15	30
15 – 20	35
20 – 25	40
25 – 30	**75**
30 – 35	15

2. Identify the highest frequency

Scan the frequency values $(30, 35, 40, 75, 15)$ to find the maximum value:

$\text{Maximum Frequency} = 75$

3. Determine the corresponding class interval

The class interval that aligns with the maximum frequency of 75 is $25 - 30$. In statistics, the group containing the highest occurrence rate is defined as the modal class.

The modal class for the given distribution is $25 - 30$.

Q3 Marks of 60 students in a class are tabulated below. For finding mode the Lower limit of modal class (L) = -------

Marks	10 – 19	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69
Number of students	3	8	16	12	10	8

ANSWER / SOLUTION

1. Identify the modal class

The modal class is the class interval with the highest frequency.

The maximum frequency is 16. This corresponds to the class interval $30 - 39$.

2. Convert to continuous class intervals

To find the exact value of L for the standard mode formula, the data must be continuous.

$\text{Adjustment factor} = \frac{\text{Lower limit of next class} - \text{Upper limit of current class}}{2} = \frac{20-19}{2} = 0.5$

Subtract 0.5 from each lower limit and add 0.5 to each upper limit:

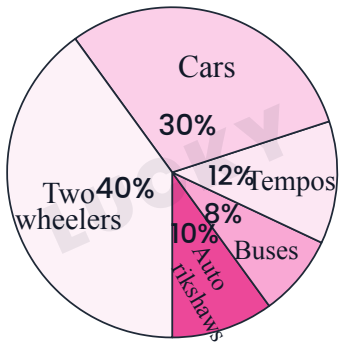
Marks (Original)	Marks (Continuous)	Number of students (Frequency)
10 – 19	9.5 – 19.5	3
20 – 29	19.5 – 29.5	8
30 – 39	**29.5 – 39.5**	**16 (Highest)**
40 – 49	39.5 – 49.5	12
50 – 59	49.5 – 59.5	10
60 – 69	59.5 – 69.5	8

3. Determine the Lower Limit (L)

In the continuous form, the modal class is $29.5 - 39.5$.

Therefore, the precise lower boundary (L) used in the mode formula is 29.5.

Q4 Observe the adjacent pie diagram. It shows the percentages of number of vehicles passing a signal in a town between 8 am and 10 am. Find central angle for cars.



TEXTBOOK Miscellaneous Problems 6 · Q.14 (1) · Page 167

ANSWER / SOLUTION

Given: Cars $= 30\%$ of the pie (percentage from the diagram).

$\text{Central angle} = \frac{\text{Percentage}}{100} \times 360^\circ = \frac{30}{100} \times 360^\circ = 0.30 \times 360^\circ = 108^\circ$

The central angle for cars is 108°.

Q5 The table given below gives the marks of 160 students in the school.

Marks	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100
No. of students	8	10	15	25	30	38	24	10

How many students are getting above 80 marks.

ANSWER / SOLUTION

To find the number of students getting above 80 marks, we need to look at the class intervals that represent marks greater than 80.

These intervals are: **80 – 90** marks and **90 – 100** marks.

From the given table:

Number of students scoring 80 – 90 marks $= 24$

Number of students scoring 90 – 100 marks $= 10$

$\text{Total students} = 24 + 10 = 34$

Q6 If the mean of the frequency distribution is 8.1 and $\sum f_i x_i = 132 + 5k$, $\sum f_i = 20$ then find k.

ANSWER / SOLUTION

1. State the mean formula

The mean ($\bar{x}$) of a grouped frequency distribution is calculated using the formula:

$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

2. Substitute given values

$8.1 = \frac{132+5k}{20}$

3. Solve for k

$8.1 \times 20 = 132 + 5k$

$162 = 132 + 5k$

$162 - 132 = 5k \implies 30 = 5k \implies k = \frac{30}{5} = 6$

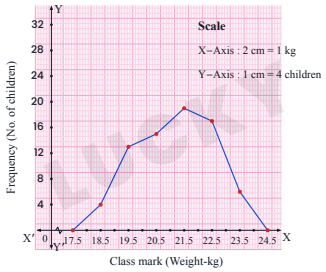
Q.2 (A) Activities

2 Marks each

4 QUESTIONS · 2 Revised

Q1 Observe the frequency polygon and complete the table.

Class	18 – 19	19 – 20	20 – 21	21 – 22	22 – 23	23 – 24
Class mark	18.5	▭	20.5	21.5	▭	23.5
Frequency	4	13	15	19	17	6
Coordinates of points	(18.5, 4)	(19.5, 13)	▭	(21.5, 19)	▭	(23.5, 6)



ANSWER / SOLUTION

For class 19 – 20: Class mark: $\frac{19+20}{2} = 19.5$

For class 20 – 21: Coordinates of points: $(20.5, 15)$

For class 22 – 23: Class mark: $\frac{22+23}{2} = 22.5$, Coordinates of points: $(22.5, 17)$

Class	18 – 19	19 – 20	20 – 21	21 – 22	22 – 23	23 – 24
Class mark	18.5	$\boxed{19.5}$	20.5	21.5	$\boxed{22.5}$	23.5
Frequency	4	13	15	19	17	6
Coordinates of points	(18.5, 4)	(19.5, 13)	$\boxed{(20.5, 15)}$	(21.5, 19)	$\boxed{(22.5, 17)}$	(23.5, 6)

Q2 **Complete the following table to find the mean.**

Weekly wages (Rupees)	Class mark (x_i)	No. of workers (f_i)	f_ix_i
0 – 2000	1000	15	15000
2000 – 4000	▭	35	105000
4000 – 6000	5000	50	250000
6000 – 8000	7000	20	140000
		$N = 120$	$\sum f_ix_i =$ ▭

$\text{Mean} = \frac{\sum f_ix_i}{\sum f_i}$

$= \frac{▭}{120}$

$=$ ▭

TEXTBOOK **Miscellaneous Problems 6 · Q.4 · Page 166**

ANSWER / SOLUTION

Weekly wages (Rupees)	Class mark (x_i)	No. of workers (f_i)	f_ix_i
0 – 2000	1000	15	15000
2000 – 4000	$\boxed{3000}$	35	105000
4000 – 6000	5000	50	250000
6000 – 8000	7000	20	140000
		$N = 120$	$\sum f_ix_i = \boxed{510000}$

$\text{Mean} = \frac{\sum f_ix_i}{\sum f_i}$

$= \frac{\boxed{510000}}{120} = \boxed{4250}$

Q3 **The following table shows causes of noise pollution. Complete the table to draw a pie diagram.**

Causes	Percentage	Central angle
Construction	10%	▭
Traffic	▭	180°
Aircraft take off	9%	▭
Industry	20%	▭
Trains	11%	39.6°

TEXTBOOK **Miscellaneous Problems 6 · Q.15 · Page 167**

ANSWER / SOLUTION

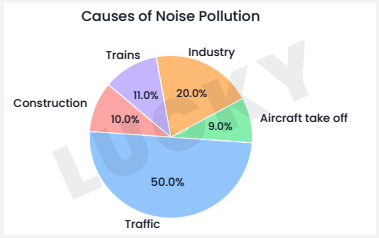
1. Construction: Given percentage $= 10\%$. Central angle $= \frac{10}{100} \times 360^\circ = 36^\circ$

2. Traffic: Given central angle $= 180^\circ$. Percentage $= \frac{180^\circ}{360^\circ} \times 100 = 50\%$

3. Aircraft take off: Given percentage $= 9\%$. Central angle $= \frac{9}{100} \times 360^\circ = 32.4^\circ$

4. Industry: Given percentage $= 20\%$. Central angle $= \frac{20}{100} \times 360^\circ = 72^\circ$

Causes	Percentage	Central angle
Construction	10%	$\boxed{36^\circ}$
Traffic	$\boxed{50\%}$	180°
Aircraft take off	9%	$\boxed{32.4^\circ}$
Industry	20%	$\boxed{72^\circ}$
Trains	11%	39.6°



Q4 **Complete the following activity to find median.**

Class (Student's marks)	No. of students (f_i)	Cumulative frequency less than the upper limit (cf)
0 – 20	4	4
20 – 40	20	24
40 – 60	30	54
60 – 80	40	94
80 – 100	6	100

Here, $L =$ ▭, $N = 100$, $\frac{N}{2} = 50$, $c.f. = 24$, $f = 30$, $h = 20$

Median $=$ ▭ ---(Formula)

$=$ ▭

Median marks $=$ ▭

ANSWER / SOLUTION

Step 1: Find the missing parameters

From the given data, since $\frac{N}{2} = 50$, the cumulative frequency just greater than 50 is 54.

This corresponds to the median class of 40 – 60.

L (Lower limit of the median class) $= \boxed{40}$

Q4 *continued...*

ANSWER / SOLUTION (CONTD.)

Step 2: Formula and Calculation

$$\text{Median} = \boxed{L + \left[\frac{\frac{N}{2} - c.f.}{f}\right] \times h}$$

Substituting values: $\text{Median} = \boxed{40 + \left[\frac{50 - 24}{30}\right] \times 20}$

$\text{Median} = 40 + \left[\frac{26}{30}\right] \times 20 = 40 + \frac{52}{3} = 40 + 17.33 = \boxed{57.33}$

Q.2 (B) Short Solve

2 Marks each | 3 QUESTIONS | ☐ Revised

Q1 **For the table given below find cumulative frequency table.**

Daily No. of hours	8 – 10	10 – 12	12 – 14	14 – 16
Number of workers	150	500	300	50

ANSWER / SOLUTION

Daily No. of Hours	Number of workers (Frequency, f)	Cumulative Frequency (cf)	Calculation
8 – 10	150	150	150
10 – 12	500	650	$150 + 500$
12 – 14	300	950	$650 + 300$
14 – 16	50	1000	$950 + 50$

Q2 **Find the mode for the following frequency distribution.**

Class	1 – 10	11 – 20	21 – 30	31 – 40	41 – 50
Frequency	2	3	5	7	1

Use formula: $\text{Mode} = L + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h$

ANSWER / SOLUTION

Step 1: Convert discontinuous classes to continuous classes

Original Class	Continuous Class	Frequency
1 – 10	0.5 – 10.5	2
11 – 20	10.5 – 20.5	3
21 – 30	20.5 – 30.5	5
31 – 40	**30.5 – 40.5**	**7**
41 – 50	40.5 – 50.5	1

Step 2: Identify variables for formula

Modal class $= 30.5 - 40.5$, $L = 30.5$, $f_0 = 7$ (frequency of modal class), $f_1 = 5$ (preceding), $f_2 = 1$ (succeeding), $h = 10$.

Step 3: Calculate the mode

$\text{Mode} = L + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h$

$\text{Mode} = 30.5 + \left(\frac{7-5}{2(7)-5-1}\right) \times 10 = 30.5 + \left(\frac{2}{14-6}\right) \times 10 = 30.5 + \frac{2}{8} \times 10 = 30.5 + 2.5 = 33$

Q3 **Find mean of the following table.**

Difference in ages (in years)	No. of couples (f_i)
0 – 2	1
2 – 4	2
4 – 6	8
6 – 8	5
8 – 10	3
10 – 12	1

ANSWER / SOLUTION

1. Calculation table

Class Mark $(x_i) = \frac{\text{Lower Limit}+\text{Upper Limit}}{2}$

Difference in ages (in years)	No. of couples (f_i)	Class Mark (x_i)	$f_i \times x_i$
0 – 2	1	1	$1 \times 1 = 1$
2 – 4	2	3	$2 \times 3 = 6$
4 – 6	8	5	$8 \times 5 = 40$
6 – 8	5	7	$5 \times 7 = 35$
8 – 10	3	9	$3 \times 9 = 27$
10 – 12	1	11	$1 \times 11 = 11$
Total	$\sum f_i = 20$		$\sum f_i x_i = 120$

2. Formula and Final Steps

$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = \frac{120}{20} = 6$

Q.3 (A) Activities 3 Marks each

4 QUESTIONS · 1 ☐ Revised

Q1 The following table shows the funds collected by 50 students for flood affected people. Find the mean of the funds.

Fund (Rupees)	0 – 1000	1000 – 1500	1500 – 2000	2000 – 3000
No. of students	6	24	18	2

Solution: Let $A = 1250$, examining all $d_i, g = 250$.

Class Fund (₹)	Class mark x_i	$d_i = x_i - A = x_i - 1250$	$u_i = \frac{d_i}{g}$	Frequency f_i	f_iu_i
0 – 1000	500	▭	–3	6	–18
1000 – 1500	$1250 \to A$	0	0	24	0
1500 – 2000	1750	500	2	18	▭
2000 – 3000	2500	1250	▭	2	10
Total				$\sum f_i = 50$	$\sum f_iu_i = 28$

$\bar{u} = \frac{\sum f_iu_i}{\sum f_i} = \frac{28}{50} =$ ▭

$\bar{u}g =$ ▭ $\times 250 =$ ▭

$\bar{x} = A + g\bar{u} = 1250 + 140 = 1390$

∴ The average of the funds is ₹ 1390.

TEXTBOOK Solved Example · Ex. (1) · Page 137

ANSWER / SOLUTION

Let $A = 1250$, examining all $d_i, g = 250$.

Class Fund (₹)	Class mark x_i	$d_i = x_i - A = x_i - 1250$	$u_i = \frac{d_i}{g}$	Frequency f_i	f_iu_i
0 – 1000	500	$\boxed{-750}$	–3	6	–18
1000 – 1500	$1250 \to A$	0	0	24	0
1500 – 2000	1750	500	2	18	$\boxed{36}$
2000 – 3000	2500	1250	$\boxed{5}$	2	10
Total				$\sum f_i = 50$	$\sum f_iu_i = 28$

$\bar{u} = \frac{\sum f_iu_i}{\sum f_i} = \frac{28}{50} = \boxed{0.56}$

$\bar{u}g = \boxed{0.56} \times 250 = \boxed{140}$

$\bar{x} = A + g\bar{u} = 1250 + 140 = 1390$

∴ The average of the funds is ₹ 1390.

Q2 **Complete the following activity to find mean by direct method.**

Weekly wages (Rupees)	Class mark (x_i)	No. of workers (f_i)	$x_i f_i$
1000 – 2000	1500	25	37500
2000 – 3000	▭	45	112500
3000 – 4000	3500	50	▭
4000 – 5000	4500	30	▭
		$N = 150$	$\sum f_i x_i =$ ▭

Mean $= \frac{\sum f_i x_i}{\sum f_i}$

$= \frac{▭}{150}$

$=$ ▭

✓ ANSWER / SOLUTION

Weekly wages (Rupees)	Class mark (x_i)	No. of workers (f_i)	$x_i f_i$
1000 – 2000	1500	25	37500
2000 – 3000	$\boxed{2500}$	45	112500
3000 – 4000	3500	50	$\boxed{175000}$
4000 – 5000	4500	30	$\boxed{135000}$
		$N = 150$	$\sum f_i x_i = \boxed{460000}$

Mean $= \frac{\sum f_i x_i}{\sum f_i}$

$= \frac{\boxed{460000}}{150} = \boxed{3066.67}$

Q3 **Complete the following activity to find median marks of the students.**

Class (Student's marks)	No. of students (f_i)	Cumulative frequency less than the upper limit (cf)
0 – 20	4	4
20 – 40	20	24
40 – 60	30	54
60 – 80	40	94
80 – 100	6	100

Here, $L =$ ▭, $N = 100$, $\frac{N}{2} = 50$, $c.f. =$ ▭, $f = 30$, $h = 20$

Median $= L + \frac{\frac{N}{2} - c.f.}{f} \times h$ ---(Formula)

$=$ ▭ ---(Keeping values in the formula)

$=$ ▭ $+ \frac{▭}{30} \times 20$

Median marks $=$ ▭

✓ ANSWER / SOLUTION

Here, $L = \boxed{40}$, $N = 100$, $\frac{N}{2} = 50$, $c.f. = \boxed{24}$, $f = 30$, $h = 20$

Median $= L + \frac{\frac{N}{2} - c.f.}{f} \times h$ ---(Formula)

$= \boxed{40 + \frac{50 - 24}{30} \times 20}$ ---(Keeping values in the formula)

$= \boxed{40} + \frac{\boxed{26}}{30} \times 20$

$= 40 + \frac{52}{3} = 40 + 17.33$

Median marks $= \boxed{57.33}$

Q4 **The table below gives the ages of 120 drivers of cars involved in accidents during a certain year. Complete the table and draw a pie diagram.**

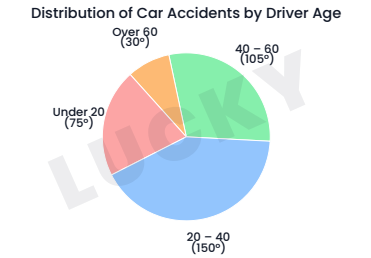
Age of driver (in years)	Number of drivers	Central angle
Under 20	25	75°
20 – 40	50	▭
40 – 60	35	105°
Over 60	10	30°

ANSWER / SOLUTION

1. Completed table

Age of driver (in years)	Number of drivers	Central angle
Under 20	25	75°
20 – 40	50	$\boxed{150^\circ}$
40 – 60	35	105°
Over 60	10	30°
Total	120	360°

2. Pie diagram



Q.3 (B) Solve / Prove

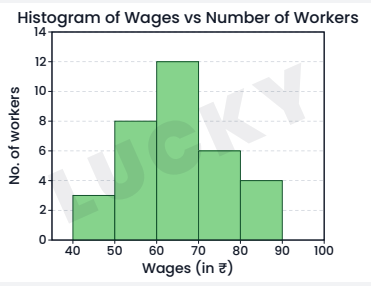
3 Marks each

3 QUESTIONS ☐ Revised

Q1 **Draw the histogram.**

Wages (in ₹)	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
No. of workers	3	8	12	6	4

ANSWER / SOLUTION



Q2 **In a school the weekly pocket money of 50 students is as follows. Find mode of pocket money.**

Pocket money (in ₹)	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100
No. of students	2	8	12	14	8	6

ANSWER / SOLUTION

1. Identify the modal class

The modal class is the class interval with the highest frequency.

The maximum frequency is 14, which belongs to the class interval 70 – 80.

Therefore, the modal class $= 70 - 80$.

2. Extract the Values

l (lower limit of the modal class) $= 70$

f_1 (frequency of the modal class) $= 14$

f_0 (frequency of the preceding class) $= 12$

f_2 (frequency of the succeeding class) $= 8$

h (class size or width) $= 80 - 70 = 10$

3. Substitute values into the formula

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 70 + \left(\frac{14-12}{2(14)-12-8}\right) \times 10 = 70 + \left(\frac{2}{28-20}\right) \times 10 = 70 + \frac{2}{8} \times 10 = 70 + 2.5 = 72.5$$

Q3 **Draw frequency polygon of the following distribution.**

Weight of packet	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
Number of packets	3	5	8	5	4

ANSWER / SOLUTION

1. Class mark calculation table

Weight of Packet (Class Interval)	Class Mark (x-axis)	Number of Packets (Frequency, y-axis)	Plotting Point (x, y)
–5 – 0 (Imaginary)	–2.5	0	(–2.5, 0)
0 – 5	2.5	3	(2.5, 3)
5 – 10	7.5	5	(7.5, 5)
10 – 15	12.5	8	(12.5, 8)
15 – 20	17.5	5	(17.5, 5)
20 – 25	22.5	4	(22.5, 4)
25 – 30 (Imaginary)	27.5	0	(27.5, 0)

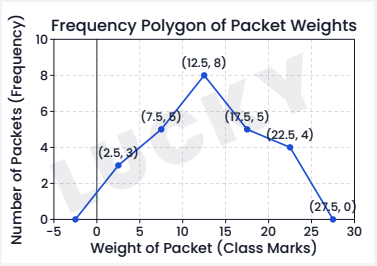
2. Frequency polygon graph

3. Steps to draw it on graph paper:

Axes Setup: Plot weight of packet on the horizontal x-axis and number of packets on the vertical y-axis.

Plot Points: Mark the coordinates (x, y) from the table above as dots on your graph paper.

Connect Points: Use a straight ruler to join the consecutive dots one by one.



Q.4 Long Solve / Prove — 4 Marks each

5 QUESTIONS ☐ Revised

Q1 **Find the mean by step deviation method.**

Class	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35	35 – 40
Frequency	5	6	8	12	6	3

ANSWER / SOLUTION

1. Formula

The formula for finding the mean using the step-deviation method is:

$\text{Mean } (\bar{x}) = A + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h$

Where: A = Assumed mean (middle value from class marks x_i), h = Class width (upper limit $-$ lower limit $= 15 - 10 = 5$), $u_i = \frac{x_i - A}{h}$, x_i = Class mark $= \frac{\text{Lower limit} + \text{Upper limit}}{2}$

2. Calculation table

Assumed mean $A = 22.5$ and class width $h = 5$:

Class	Frequency (f_i)	Class Mark (x_i)	$u_i = \frac{x_i - 22.5}{5}$	$f_i u_i$
10 – 15	5	12.5	–2	–10
15 – 20	6	17.5	–1	–6
20 – 25	8	22.5 (A)	0	0
25 – 30	12	27.5	1	12
30 – 35	6	32.5	2	12
35 – 40	3	37.5	3	9
Total	$\sum f_i = 40$			$\sum f_i u_i = 17$

3. Substitute values into the formula

$\bar{x} = 22.5 + \left(\frac{17}{40}\right) \times 5 = 22.5 + \frac{17}{8} = 22.5 + 2.125 = 24.625$

Q2 **If the mean of the weight of packet is 12.9 gm. Find the value of b.**

Weight of packet	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
Number of packets	3	b	8	5	4

ANSWER / SOLUTION

1. Construct the frequency table

Weight of packet (Class Interval)	Number of packets (f_i)	Class Midpoint (x_i)	$f_i \times x_i$
0 – 5	3	2.5	$3 \times 2.5 = 7.5$
5 – 10	b	7.5	$7.5b$
10 – 15	8	12.5	$8 \times 12.5 = 100$
15 – 20	5	17.5	$5 \times 17.5 = 87.5$
20 – 25	4	22.5	$4 \times 22.5 = 90$
Total	$\sum f_i = 20 + b$		$\sum f_i x_i = 285 + 7.5b$

2. Substitute into the mean formula

Given that Mean $(\bar{x}) = 12.9$:

$12.9 = \frac{285 + 7.5b}{20 + b}$

$12.9 \times (20 + b) = 285 + 7.5b \implies 258 + 12.9b = 285 + 7.5b$

$12.9b - 7.5b = 285 - 258 \implies 5.4b = 27 \implies b = \frac{27}{5.4} = 5$

Q3 **Find the mode for the following distribution.**

Ages (in years)	1 – 10	11 – 20	21 – 30	31 – 40	41 – 50
No. of children	2	3	5	7	1

✎ CORRECTED ~~30.5 + 2.5 = 33.5~~ → **30.5 + 2.5 = 33** — addition error (30.5 + 2.5 = 33; same data in Q.2 (B) Q2 gives 33)

✓ **ANSWER / SOLUTION**

1. Make class boundaries continuous

Age Interval (Inclusive)	Continuous Class Boundaries	No. of Children (f)
1 – 10	0.5 – 10.5	2
11 – 20	10.5 – 20.5	3
21 – 30	20.5 – 30.5	5
31 – 40	30.5 – 40.5	7 (f_1)
41 – 50	40.5 – 50.5	1

2. Identify modal class variables

$L = 30.5,\ f_1 = 7,\ f_0 = 5,\ f_2 = 1,\ h = 10$

3. Compute using mode formula

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 30.5 + \left(\frac{7-5}{2(7)-5-1}\right) \times 10 = 30.5 + \frac{2}{8} \times 10 = 30.5 + 2.5 = 33$$

Q4 **Calculate the missing frequency from the following distribution, it being given that the median of the distribution is 24.**

Age in years	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of persons	5	25	?	18	7

✓ **ANSWER / SOLUTION**

Class interval	Frequency	Cumulative frequency
0 – 10	5	5
10 – 20	25	30
20 – 30	x	$30 + x$
30 – 40	18	$48 + x$
40 – 50	7	$55 + x$
	$N = 55 + x$	

Given: $\text{Median} = 24$. Then median class $= 20 - 30$.

$l = 20,\ h = 30 - 20 = 10,\ F = 30$ and $f = x$

$$\text{Median} = l + \left(\frac{\frac{N}{2} - F}{f}\right) \times h$$

$$24 = 20 + \left(\frac{\frac{55+x}{2} - 30}{x}\right) \times 10 \implies 4x = \left(\frac{55+x-60}{2}\right) \times 10$$

$$8x = (x - 5) \times 10 \implies 8x = 10x - 50 \implies 2x = 50 \implies x = 25$$

$\therefore$ Missing frequency $= 25$

Q5 **Draw histogram and frequency polygon.**

Max. temp.	24 – 28	28 – 32	32 – 36	36 – 40	40 – 44
No. of towns	4	5	7	8	6

ANSWER / SOLUTION

1. Prepare data table

Class Mark $= \frac{\text{Lower Limit}+\text{Upper Limit}}{2}$

Additionally, create two imaginary classes with frequency 0 at both ends ($20-24$ and $44-48$) to anchor the polygon to the horizontal axis.

Max.Temperature (°C)	Class Mark (Midpoint)	No. of Towns (Frequency)	Points for Polygon (x, y)
20 – 24 (Imaginary)	22	0	(22, 0)
24 – 28	26	4	(26, 4)
28 – 32	30	5	(30, 5)
32 – 36	34	7	(34, 7)
36 – 40	38	8	(38, 8)
40 – 44	42	6	(42, 6)
44 – 48 (Imaginary)	46	0	(46, 0)

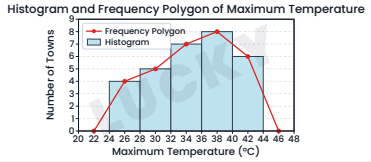
2. Draw histogram

3. Plot frequency polygon

Identify coordinates: Locate the center top point of each histogram bar, which corresponds to coordinates $(ClassMark, Frequency)$.

Connect the points: Join consecutive points with straight line segments.

Close the polygon: Extend lines to imaginary points $(22, 0)$ and $(46, 0)$ on the X-axis.



Q.5 Creative / HOTS 3 Marks each

3 QUESTIONS ☐ Revised

Q1 **The angles of a triangle ABC are in the ratio $1:2:3$, show this information in pie diagram.**

ANSWER / SOLUTION

1. Find the angles of the triangle

Interior angles add up to $180°$. Given ratio $1:2:3$, let the angles be $x, 2x$ and $3x$.

$x + 2x + 3x = 180° \implies 6x = 180° \implies x = 30°$

$\angle A = 30°, \angle B = 60°, \angle C = 90°$

2. Calculate the central angles for the pie diagram

Central angle $= \frac{\text{Component value}}{\text{Total value}} \times 360°$

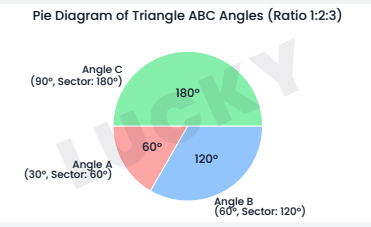
Section A ($\angle A = 30°$): $\frac{30°}{180°} \times 360° = 60°$

Section B ($\angle B = 60°$): $\frac{60°}{180°} \times 360° = 120°$

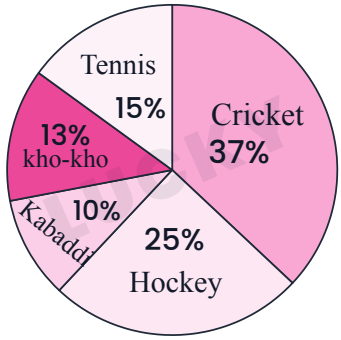
Section C ($\angle C = 90°$): $\frac{90°}{180°} \times 360° = 180°$

3. Visual representation (Pie diagram)

The angles of triangle ABC map onto a pie diagram with sector central angles of $60°$, $120°$ and $180°$ respectively.



Q2 120 students of standard 10 were asked which game they like and from this information pie chart is prepared, observe the pie diagram and answer the questions.



a. How many students like to play Tennis?
b. What will be central angle for Kho-Kho?
c. How many students like Kabaddi?

ANSWER / SOLUTION

Given: 120 students and pie-chart percentages: Tennis $= 15\%$, Kho-Kho $= 13\%$, Kabaddi $= 10\%$.

Step-wise calculation:

a. Number who like Tennis:

15% of $120 = \frac{15}{100} \times 120 = 0.15 \times 120 = 18$

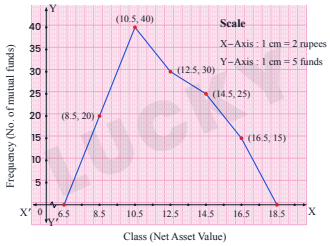
b. Central angle for Kho-Kho:

$\text{Central angle} = \frac{\text{Percentage}}{100} \times 360^\circ = \frac{13}{100} \times 360^\circ = 46.8^\circ \quad (46^\circ 48')$

c. Number who like Kabaddi:

10% of $120 = \frac{10}{100} \times 120 = 0.10 \times 120 = 12$

Q3 Observe the frequency polygon given alongside. Prepare frequency table and find mean of the net asset value.



ANSWER / SOLUTION

1. Determining the class intervals

Class marks (x_i) are plotted on x-axis. Class width (g) $= 10.5 - 8.5 = 2$.

Lower limit $= x_i - \frac{g}{2} = x_i - 1$

Upper limit $= x_i + \frac{g}{2} = x_i + 1$

2. Frequency table

Class (Net Asset Value)	Class Mark (x_i)	Frequency (No. of funds, f_i)	Product ($f_i \times x_i$)
7.5 – 9.5	8.5	20	170
9.5 – 11.5	10.5	40	420
11.5 – 13.5	12.5	30	375
13.5 – 15.5	14.5	25	362.5
15.5 – 17.5	16.5	15	247.5
Total		$\sum f_i = 130$	$\sum f_i x_i = 1575$

3. Calculating the mean

$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = \frac{1575}{130} \approx 12.1154 \approx 12.12$

Thus, the mean of the net asset value is approximately 12.12.

Practice daily. Solve every step. Score full marks.
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